

Worked Example: Instantiating a Tiered Reader-Privacy Receipt

Receipt line items, user discovery, and evaluation interlocks

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Abstract

The synthesis layer provides clean interfaces for receipt line items, user-verifiability, threat windows, exposure normal forms, and auditor replay hooks. This note is a single end-to-end instantiation whose job is *coherence*, not novelty: a reader-privacy claim with a relayed delegated-routing primary path and a budgeted structured-overlay fallback. It is an “integration test” for the publication surface: a referee should be able to understand *what is being claimed*, a user should be able to *diff the claim across releases*, and an auditor should be able to *replay* the declared computations after requesting the private artifact bundle.

How to read this note. Skip everything except: (i) Section 1 for the declared threat window and exposure surfaces, (ii) Section 2 for the five representative receipt line items and their replay hooks, and (iii) Section 6 for the tiny arithmetic slice that downstream tools consume. All other sections are there only to show the interfaces compose without “semantic drift”.

Owned background (avoid redundancy). Schema-level receipt tuples and JSONPath meanings are owned by Synthesis 12.[4] Exposure normal forms (ENF IDs) and observation tiers are owned by Synthesis 3 and Synthesis 15.[6, 17] Threat windows (TW IDs) are owned by Synthesis 4.[7] Primary-path separation guards are owned by Synthesis 22–23.[11, 12] Replay-plan minimality and plan/state equivalence are owned by Synthesis 18–20.[14, 29] User discovery and minimal interfaces are owned by Synthesis 13.[15] Endpoint conversions are owned by Synthesis 5 (and odds inflation by Anonymity A).[8, 9] Artifact availability and disclosure are governed by ABOM/OINL and the series artifact policy.[18, 2]

1 Scenario and declared interfaces

We instantiate one deployment slice with two paths:

1. **Primary: relayed delegated routing.** A relay separates client-side metadata from destination-side metadata (a primary separation claim guarded by a published failure probability ρ). Deployed control-plane surfaces for delegated routing (AutoConf, overrides, path scopes) are pinned by the state declaration per Synthesis 30.[11, 3]
2. **Fallback: structured overlay lookup.** When primary fails (or is bypassed by policy), a structured overlay lookup is executed and may expose a Tier 2 committee-contact trace (CCT) together with coarse timing and congestion witnesses.[6, 17]

Public object (digest-bound)	What it fixes (why a user can diff it)
<code>receipt_id</code>	The line items: declared interfaces, budgets, and replay hooks (Synthesis 12).
<code>tw_id</code>	Threat window declaration: secret, repetition unit/window, adversary model (Synthesis 4).
<code>enf_registry_id</code>	Exposure normal form labels: which transcript projection is in scope (Synthesis 3).
<code>state_decl_id</code>	The small public state surface that conditions the claim (state series + Synthesis 18).
<code>primary_surface_manifest_id</code>	Guard object for non-collusion/separation assumptions (Synthesis 22).
<code>replay_plans_id</code>	Auditor replay recipe: how to recompute each line item from private artifacts (Synthesis 20).

Table 1: Minimal public identifiers for the worked example. The point is not the particular files but the *digest discipline*: a name like `enf.fallback.cct.v1` is not trusted unless its digest matches.

The deployer publishes a small *public bundle* (receipt + digest-bound companion objects) and retains a *private bundle* (raw logs, scripts, traces, and any privileged measurements). Users do *not* need the private bundle to understand or compare the claim; auditors can request it to replay the declared computations.[29]

Primary-path specificity without bloat. The public `primary_surface_manifest_id` is intentionally a *minimal guard object*: it commits to the separation class and the clauses that would invalidate the non-collusion assumption.[11] Protocol- and endpoint-level details (e.g., the delegated-routing API version, endpoint set, service discovery/AutoConf mode, and any caching knobs) live in the private bundle as a config snapshot referenced by the manifest’s evidence hooks; this keeps the public receipt diffable without forcing a full configuration dump into the paper. For the deployed-surface taxonomy and a “what-to-pin” checklist (projection vs state vs separation), see Synthesis 30.[3]

Where the worked bundle lives. In the archive this example is represented by the receipt file `artifacts/example_receipt.json` together with digest-bound companion objects. The full file list is intentionally not reproduced here; it is discoverable from `support_manifest.json` and `example_artifact_inventory.json` in the bundle.

2 Receipt line items

This note does not re-specify the receipt schema; it *instantiates* it. The canonical tuple schema lives in Synthesis 12, and deployed control-plane surfaces referenced by any line item are owned by Synthesis 30.[4, 3] Here we keep only the slice needed to make the scenario mechanically replayable.

Minimal slice used here.

1. **Interface IDs:** `tw_id` (threat window template) and `exposure_nf_id` (declared observable projection / ENF)—owned by Synthesis 4 and Synthesis 3.[7, 6]
2. **State binding:** `state_decl_id` pins the state surfaces this bundle admits (e.g., the public refresh timer and bucket-level staleness) and therefore what “replay” means.[14]

3. **Primary separation:** `primary_path_declaration` publishes the separation class and its failure guard $(0, \rho)$. [11, 12]
4. **Fallback exposure:** `fallback_contact_surface` and `fallback_tiered_vector` publish attenuated Tier 2 summaries under an explicit activation cap. [13, 16, 17]
5. **Repeated-use profiling:** `profiling_equalization` publishes a profiling/equalization summary that can be compared across releases. [10]

Line item	Interface ids (binding)	Budget meaning (one line)
<code>primary_path_declaration</code>	<code>tw_id, primary_surface_manifest_id</code>	Guarded non-collusion checkpoint $(0, \rho)$ for the relayed primary path.
<code>fallback_contact_surface</code>	<code>tw_id, state_decl_id, exposure_nf_id=enf.fallback.cct.v1</code>	Attenuated Tier 2 contact leakage b_{att} under an activation cap p .
<code>fallback_tiered_vector</code>	<code>tw_id, exposure_nf_id=enf.fallback.tiervec.v1</code>	Tier-vector summary (Tier 0/1/2 components) for conservative composition.
<code>path_selection_indicator</code>	<code>tw_id, policy_witness_id</code>	Publishes the fallback activation cap/estimate p and how it is justified (policy vs audit).
<code>profiling_equalization</code>	<code>tw_id, exposure_nf_id=enf.profile.retry.v1</code>	Pairwise equalization / odds-inflation budget (η, δ) for repeated-use profiling control.

Table 2: Five representative receipt line items in the worked bundle, shown at the “what-to-read” granularity. Full digests and replay hooks live in `example_receipt.json`.

Receipt excerpt (worked bundle). For compactness we show only a minimal excerpt. The shipped `example_receipt.json` carries full digests and replay hooks; here we expand `fallback_contact_surface` to show the semantic contact-surface binding (see Synthesis 12). [4]

```
{
  "claim_id": "...",
  "state_decl_id": "sha256:...",
  "tw_id": "sha256:...",
  "line_items": [
    {"name": "primary_path_declaration", ...},

    {"name": "fallback_contact_surface",
      "contact_surface_id": "sha256:...",
      "contact_surface": {
        "method": "GET",
        "url": "https://cid.contact/routing/v1/providers",
        "filter_protocols": ["transport-bitswap"],
        "timeouts": {"http_router": "30s", "routing": "60s"}
      },
      ...},

    {"name": "fallback_tiered_vector", ...},
    {"name": "path_selection_indicator", ...},
  ]
}
```

```

    {"name": "profiling_equalization", ...}
  ]
}

```

What this bundle intentionally omits. This worked example is primarily a *receipt interlock* for fallback exposure and repeated-use profiling. A full deployment that uses delegated routing (Routing V1, AutoConf, router overrides, caching, trustless HTTP retrieval) should also pin those control-plane knobs in `state_decl_id` per Synthesis 30; we keep that as an explicit extension point rather than duplicating Synthesis 30 inside this paper.[3]

3 State-surface micro-slice

State makes privacy claims fragile: caches, routing tables, congestion scores, and policy flags can condition the transcript and can themselves become witnesses.[22] The worked example therefore publishes a *minimal, diffable* state declaration (digest-bound `state_decl_id`) describing only what is needed to interpret the line items:

- which caches and retry policies are in force,
- which public congestion/load beacons are consulted by routing,
- which versioned selection policy (menu) is active.

Everything else is treated as private artifact state that must be replayable by an auditor if it affects any line item.[14, 29]

4 How the interlock is supposed to work

This example is intentionally “boring”: it is a mechanical contract.

1. **User discovery.** A user finds the claim via the UVI/minimal interface object and obtains the receipt digest.[15]
2. **User diffing.** The user compares `receipt_id`, `tw_id`, `enf_registry_id`, and `state_decl_id` across releases; any semantic change forces a digest change.[4]
3. **Auditor replay.** If challenged, the deployer provides the private artifact bundle; the auditor runs the replay plans to recompute each budget number and verify it matches the receipt.[29]
4. **Downstream consumption.** Endpoint tools consume the published per-lookup MaxL/odds parameters and (separately) the TW declaration for horizon reasoning.[8, 24]

Algorithmic skeleton (client-side, one lookup). This is intentionally high-level; concrete control-plane pinning (router set precedence, path scope, filters, caching, retrieval modes) is owned by Synthesis 30.[3]

1. **Load declared state:** resolve `state_decl_id` to an effective router set and request-shaping defaults (including any protocol filters).

2. **Bind service interfaces:** when the receipt names a service contact surface, assemble the declared `contact_surface` (method + URL + shaping defaults + time budgets) and verify `contact_surface_id` matches `sha256(JCS(contact_surface)).[1]`
3. **Attempt primary path:** query the primary service using the declared threat window and ENF; if the separation guard fails or the primary times out, record a primary failure.
4. **Gate fallback:** if fallback is taken, enforce the published activation cap (path-selection indicator) and emit only the attenuated Tier 2 vector declared in the receipt.
5. **Emit replay hooks:** store digests for the replay plan and any private artifacts (so an auditor can request the bundle and rerun the accountant).
6. **Publish receipt:** commit the tuple(s) in the release receipt so users can diff across versions.

5 What changes across a release (micro-diffs)

This worked bundle is meant to make *semantic drift mechanically visible*. Three common drift patterns illustrate what should (and should not) change in a receipt:

- **Knob drift (same interface).** If the transcript projection is unchanged but a retry/timeout/width knob changes, the line item keeps the same `exposure_nf_id` but changes `knobs` and (usually) `budget_value`. Tag `lineage=same-enf` and require replay/evidence for the new number.[4, 14]
- **State-surface drift (same mechanism, different reader set).** If a deployment silently changes which delegated-routing endpoints are contacted (AutoConf defaults, cache layers, gateway exposure), the `state_decl_id` must change even if the budget arithmetic is identical. Otherwise the claim is underspecified.[3, 14]
- **Interface drift (new projection).** If the observable transcript grows (e.g., a new `/routing/v1` endpoint is used, or response streaming changes message timing), treat it as a new `exposure_nf_id` (or an explicit coarsening if the projection is strictly reduced). Do not pretend the previous certificate applies.[6, 4]

The equivalence/coarsening taxonomy for plan/state ids in Synthesis 18 is the intended shared language for these cases; this section is only the “worked example” view.[14]

User micro-diff algorithm (receipt-level). A user does not need to interpret every field; a deterministic diff is enough: (1) fetch both receipts and verify their advertised digests (`receipt_id`); (2) compare the small set of top-level interface/state ids; (3) compare line items by `name`.

Top-level: `tw_id`, `enf_registry_id`, `state_decl_id`,
`primary_surface_manifest_id`, `replay_plans_id`
 Per line item (by name): `exposure_nf_id`, `knobs`, `budget_value`,
`evidence_id`, `replay_hook`, `lineage`

Interpretation: changing `exposure_nf_id` is interface drift; changing `state_decl_id` is control-plane/reader-set drift; changing `knobs` or `budget_value` with the same `exposure_nf_id` is knob drift (same interface, new certificate).

Case	Class	What changed / required action
refresh-only-A	refresh_only	Log checkpoint / user-watch text only; rebind receipt lineage, no budget re-evaluation.
replay-required-B	replay_changed_surfaces	Fallback observation upper bound tightened ($p_{\text{obs}} : 0.02 \rightarrow 0.015$); rerun declared evaluators and publish a fresh notarized comparison.
full-recertification-C	full_recertification	Threat window and state declaration (cache trigger) changed; treat as certified-interface drift and require recertification gating.

Table 3: Three canned drift cases shipped in the worked-example bundle.

Canonicalization notes. Digest IDs should be computed over canonical bytes, not ad hoc pretty-printed JSON. For JSON objects we recommend the JSON Canonicalization Scheme (JCS, RFC 8785), i.e., define `sha256(JCS(obj)).[1]` For human micro-diffs between two receipts-as-JSON, also canonicalize *semantic sets* (sort and deduplicate arrays that represent sets, such as protocol filter lists or router allowlists) and normalize duration strings (parse `30s/0.5m` into a single unit) so reordering/formatting noise does not masquerade as drift.

Canned drift cases (bundle-provided). The shipped bundle includes `artifacts/example_drift_cases.json`, a three-row catalog of “what changed” successors for the base release label `worked-example-draft-59`. It is intended as a sanity check for the taxonomy above (refresh-only vs replay-required vs recertification).

Example output (from the shipped compare report). The bundle also includes `artifacts/example_compare` and its `user_fast_diff` contains a “first alarm” message and the minimal numeric fields that casual users usually care about. For the canned successor in that report, the fallback budget improves while `exposure_nf_id` stays fixed:

```
base:      budget_value=0.05635, total_summary_bits=0.26590
successor budget_value=0.04240, total_summary_bits=0.26518
(same exposure_nf_id; receipt digest changed)
```

In this example, *all* control-plane drift that changes the effective reader set or request shaping (endpoint sets, AutoConf expansion rules, environment overrides, protocol filters, retrieval-mode switches) is treated as **state-surface drift** and therefore must bump `state_decl_id` even if the budget arithmetic is unchanged.[3, 14]

Three tiny examples.

1. **Window drift:** keep label `tw.reader_24h.q500.v1` but change Q (or the reset rule). The digest TW changes and downstream Q -scale reasoning must be revisited.[7]
2. **Surface drift:** keep label `enf.fallback.cct.v1` but add a new Tier 2 field. The digest E_i changes; a user diff catches that the exposure surface widened.[6]

3. **Evidence drift:** keep the profiling label but change the audit projection (bucket widths) or the confidence knob. The `evidence_id` digest changes and auditors replay via the `replay_hook`. [4, 29]

Request-shaping drift (filters and budgets). If the effective delegated-routing request adds, removes, or changes request-shaping defaults or budgets (protocol filters, router allow/deny policy, path scope, routing/retrieval timeouts that govern downgrade and fallback), treat it as **state-surface drift**: it changes *who is contacted* and what termination-time behavior is feasible, and therefore can change fallback rate and exposure. Bump `state_decl_id` and treat it as recertification-triggering. [3, 14]

6 Illustrative numerical slice

This note includes exactly one numerical “shape” so a reader sees how a receipt number relates to a projection and an activation cap. All arithmetic is recomputed by the shipped validator for the maintained bundle cut (`example_validation_report.json`). [29]

Attenuation by activation probability. When a Tier 2 projection leaks b bits *conditional on fallback occurring*, and fallback occurs with probability at most p within the declared threat window, the published attenuated leakage is

$$b_{\text{att}} = \log_2((1 - p) + p2^b),$$

the standard “odds inflation” attenuation used throughout the series. [9, 13] The worked bundle instantiates this for the fallback contact surface line item; the receipt tags whether p is a policy cap or an empirical estimate.

b (bits, conditional)	p (fallback cap)	$b_{\text{att}} = \log_2((1 - p) + p2^b)$ (bits)
8	0.05	3.78
8	0.01	1.83
12	0.01	5.39

These rows are *only* intuition for how the attenuation behaves; the receipt’s authoritative numbers are rederived by the validator from the shipped bundle objects.

Why this is enough for a worked example. The details—how p is obtained (selection taxes), how b is computed for a concrete ENF (trace interfaces), and how repeated audits spend δ_{stat} —are each owned by separate synthesis notes. [16, 6, 25, 26] Keeping this section minimal avoids re-deriving results that already have their own “referee-grade” homes.

7 Optional variants (two one-paragraph receipts)

Prefix-fetch equalization projection. A variant receipt replaces the fallback ENF with a prefix-fetch projection (coarser query privacy via requesting a bucket/prefix rather than a single key). The interface change is surfaced *only* by a different ENF digest and different evidence digest; the rest of the receipt machinery is unchanged. [20, 25]

Coarsening a large-alphabet projection. A second variant receipt demonstrates safe coarsening of a large-alphabet contact projection into an audit-feasible partition. Again, the only user-visible change is via the ENF and evidence digests, plus the resulting budget number.[27] If the partition map is selected from data (rather than fixed ex ante), the selection discipline note applies: treat the map family as witness-indexed or separate pilot selection from frozen certification under change control.[28]

8 Takeaways and open edges (short)

- A referee-facing object can stay tiny: digests + a handful of line items + replay hooks.
- “Same name, different semantics” is handled by digest IDs (ENF/TW/state), not by prose.
- Notation (keys/state/transcripts, projections) is standardized in Synthesis 8; this note only instantiates it.[5]
- Guarded endpoints are essential: a primary-path claim that depends on non-collusion must ship its ρ .
- Tier vectors and selection tax compile to a single per-lookup MaxL (or odds) parameter that downstream tools can consume.[8]
- Remaining research: robust horizon accounting under stateful dependence and operator-safe drift discipline; the state series and accounting rulebook are the intended import points.[22, 24]

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